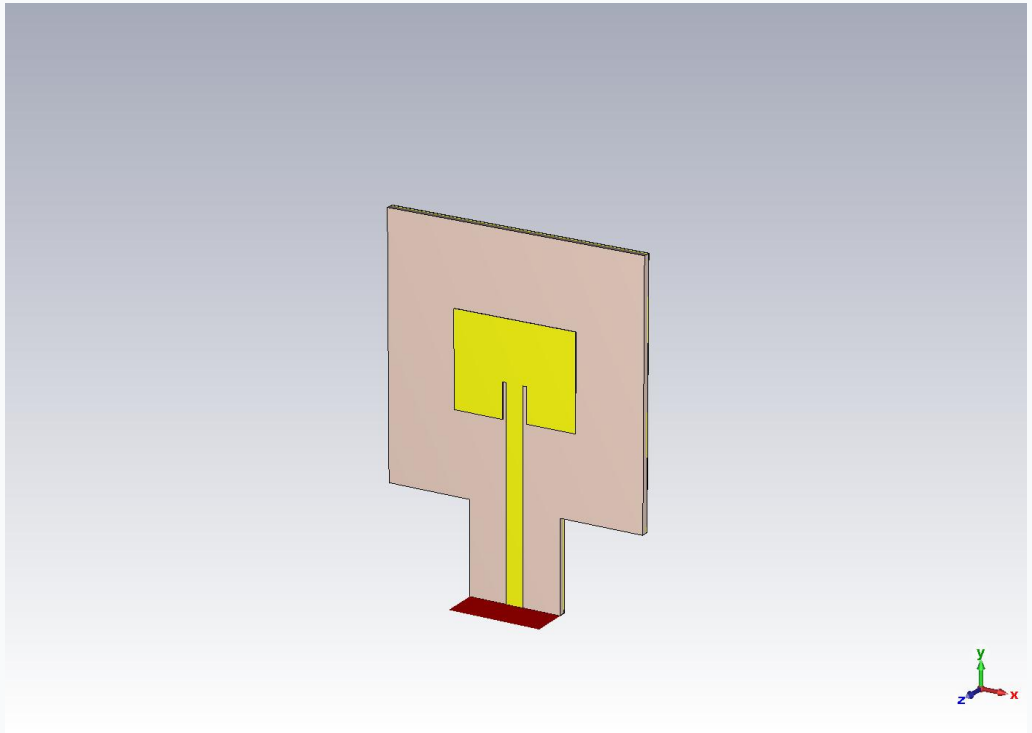
Design and Simulation of a D-Band Microstrip Patch Antenna

TELE9782

Hanqing Zhao • Ning Mao

CST Studio Suite 2024

17 November 2026



Project requirements

130–170 GHz

Specified design and sweep range

150 GHz

Nominal design point

>6.5 dBi

Maximum directivity criterion

Compare the analytical estimate and simulated match frequency

Plot S_{11} and find its −10 dB bandwidth

Plot directivity, realized gain and radiation efficiency versus frequency

Obtain the E-plane and H-plane radiation patterns

Rectangular patch theory

TM_{10} • half guided wavelength along L

$$L_{eff} \approx \lambda_0 / (2\sqrt{\varepsilon_{eff}})$$

$$L = L_{eff} - 2\Delta L$$

$$D_{max} = 4\pi U_{max} / P_{rad}$$



Plan view • schematic

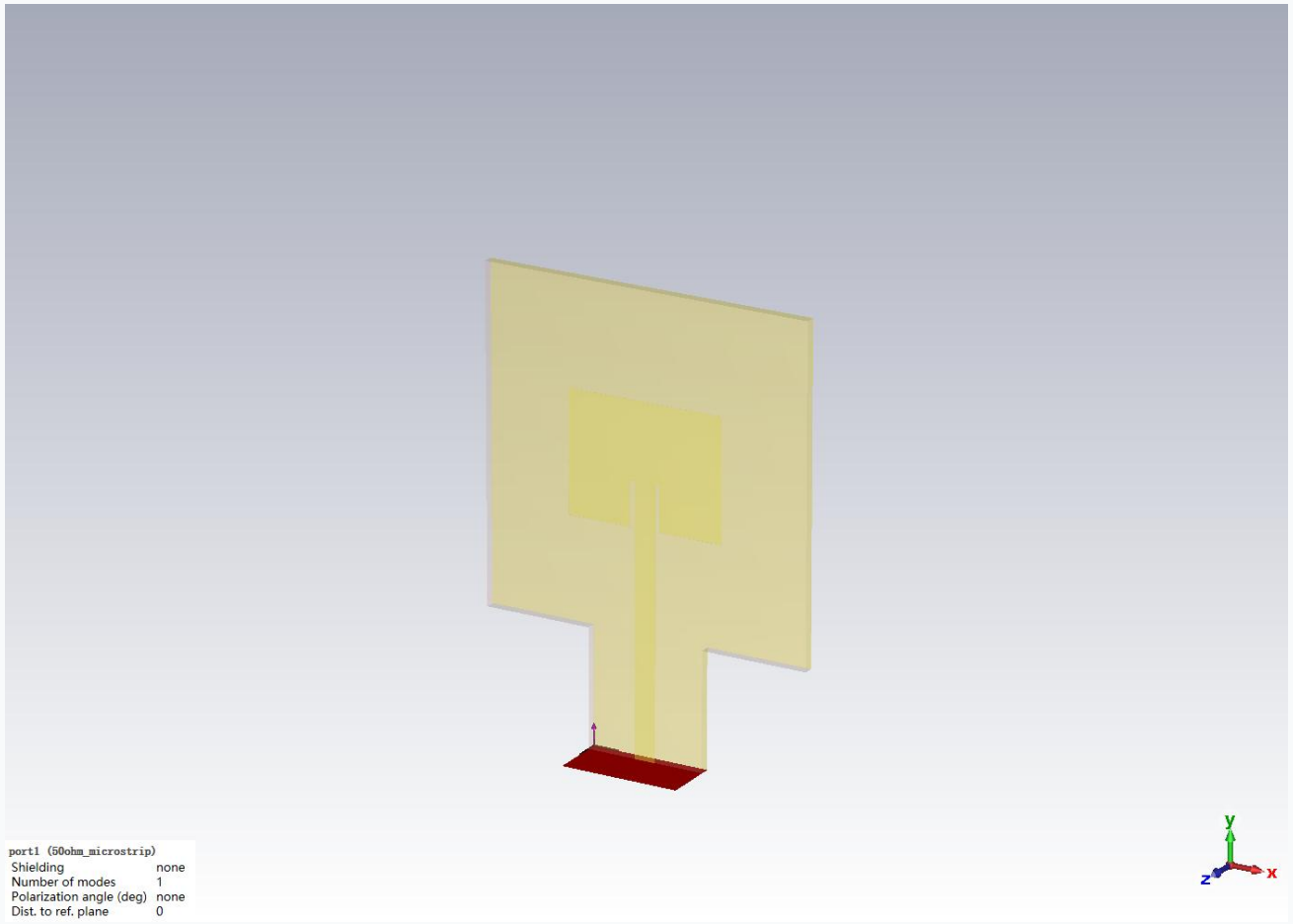
Analytical design and final parameters

Parameter	Analytical start	Final geometry
Patch width W	0.666 mm	0.667 mm
Patch length L	0.520 mm	0.513 mm
Substrate h	0.040 mm	0.040 mm
Frequency	150 GHz target	152.000 GHz analytical estimate

148.840 GHz simulated match minimum • 2.08% below the final-size estimate

Polyimide $\varepsilon r = 3.5$, $\tan\delta = 0.0027$ • Gold 0.002 mm, $\sigma = 4.561 \times 10^7$ S/m

CST model and simulation setup



0.090 mm microstrip • 50 Ω reference

Uniform launch • +y quasi-TEM mode

Open boundaries • lossy materials

Hex/PBA time domain • 130–170 GHz

262,197 cells • 13 farfield monitors

Global enclosing surface • −60 dB decay

Troubleshooting and optimisation

01 Diagnose invalid −20.92 dBi output and backward excitation

02 Rebuild a uniform launch and parameterise the finite ground

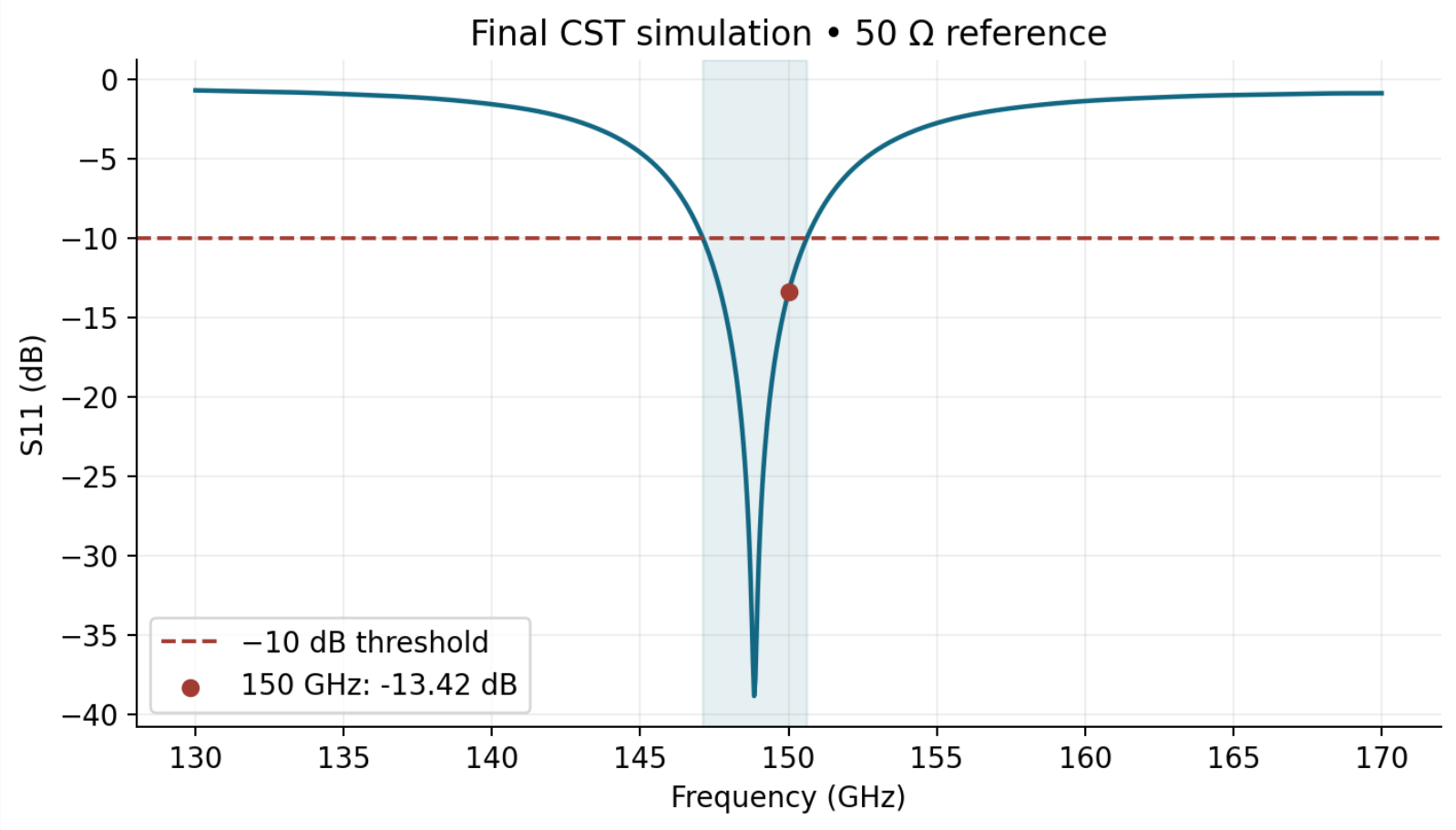
03 Baseline S_{11} min −10.18 dB at 148.56 GHz

04 Inset sweep • 0.17 mm → −14.18 dB • 0.19 mm → −26.82 dB

05 Shorten L from 0.521 to 0.513 mm to move the matched band up

06 Reject near-model farfield inconsistency; verify global-surface result

Resonance and S_{11}



150 GHz • −13.424 dB

Match dip • 148.840 GHz

Minimum • −38.859 dB

−10 dB • 147.106–150.619 GHz

Bandwidth • 3.513 GHz / 2.36%

Directivity and realized gain

7.460 dBi

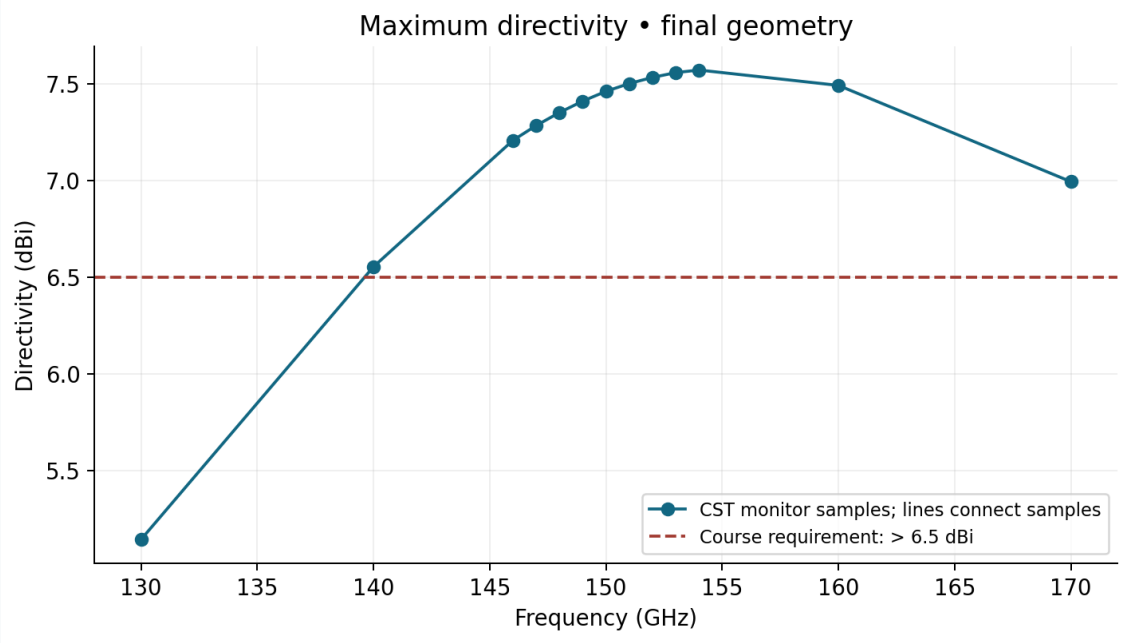
Directivity at 150 GHz

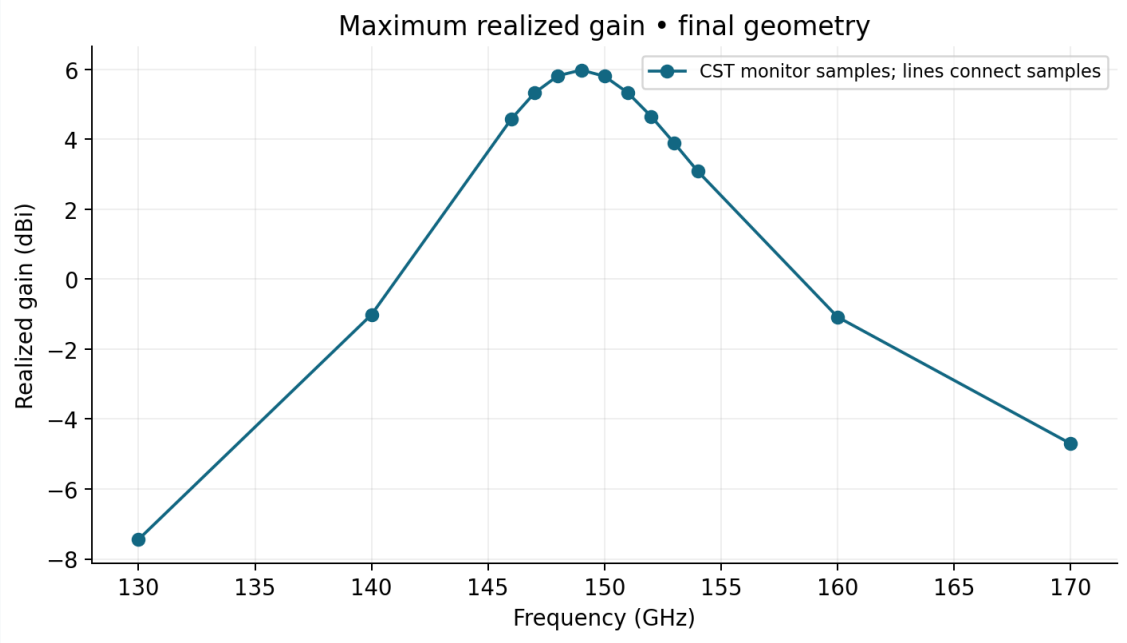
+0.960 dB

Above the 6.5 dBi criterion

5.804 dBi

Realized gain at 150 GHz

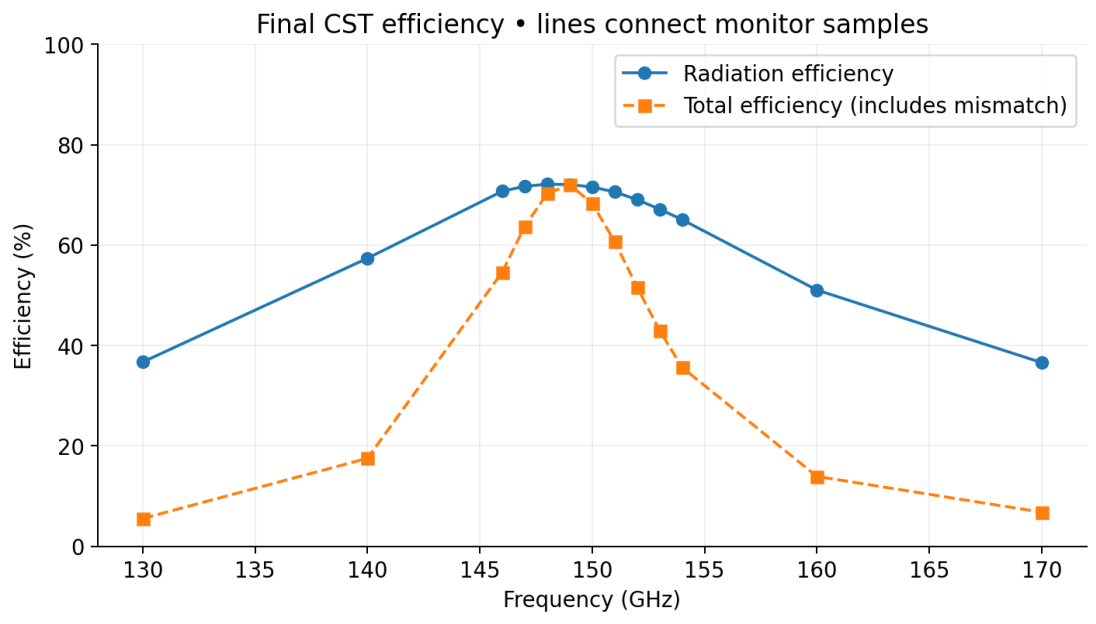




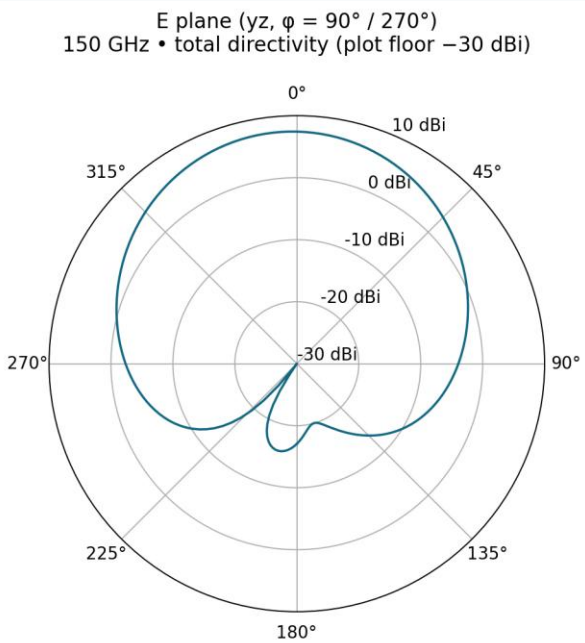
Discrete CST monitor points • realized gain includes dissipation and mismatch

Efficiency and radiation patterns

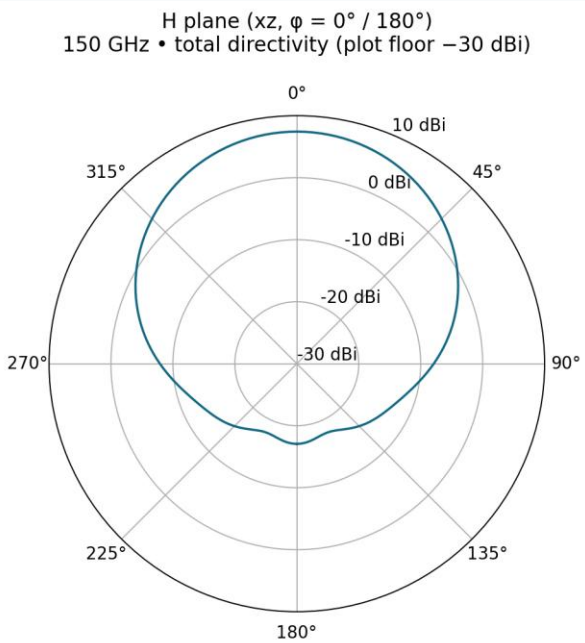
71.55% radiation efficiency at 150 GHz



E-plane / yz



H-plane / xz



E-plane yz; H-plane xz • y-polarized broadside field • peak near +z (θ ≈ 4°)

Validation and conclusion

7.460 dBi

150 GHz maximum directivity

−13.424 dB

S_{11} at the same operating point

Two meshes • ΔDmax = 0.037 dB • match dip shift 0.56%

Full sphere • mean linear D = 0.9991

Power balance • 0.754% residual of accepted power

Final CST project reopened and reran successfully

Transmission-line calculation

Start with $\lambda_0 = 299.792458 / 150 = 1.9986$ mm.

$W = \lambda_0/2 \times \sqrt{[2/(3.5 + 1)]} = 0.6662$ mm

$\varepsilon eff = (\varepsilon r + 1)/2 + (\varepsilon r - 1)/(2\sqrt{(1 + 12h/W)}) = 3.203$

$\Delta L/h = 0.412[(\varepsilon eff + 0.3)(W/h + 0.264)] / [(\varepsilon eff - 0.258)(W/h + 0.8)]$

$Leff \approx 0.5584$ mm; $\Delta L \approx 0.0190$ mm; $L \approx 0.5204$ mm

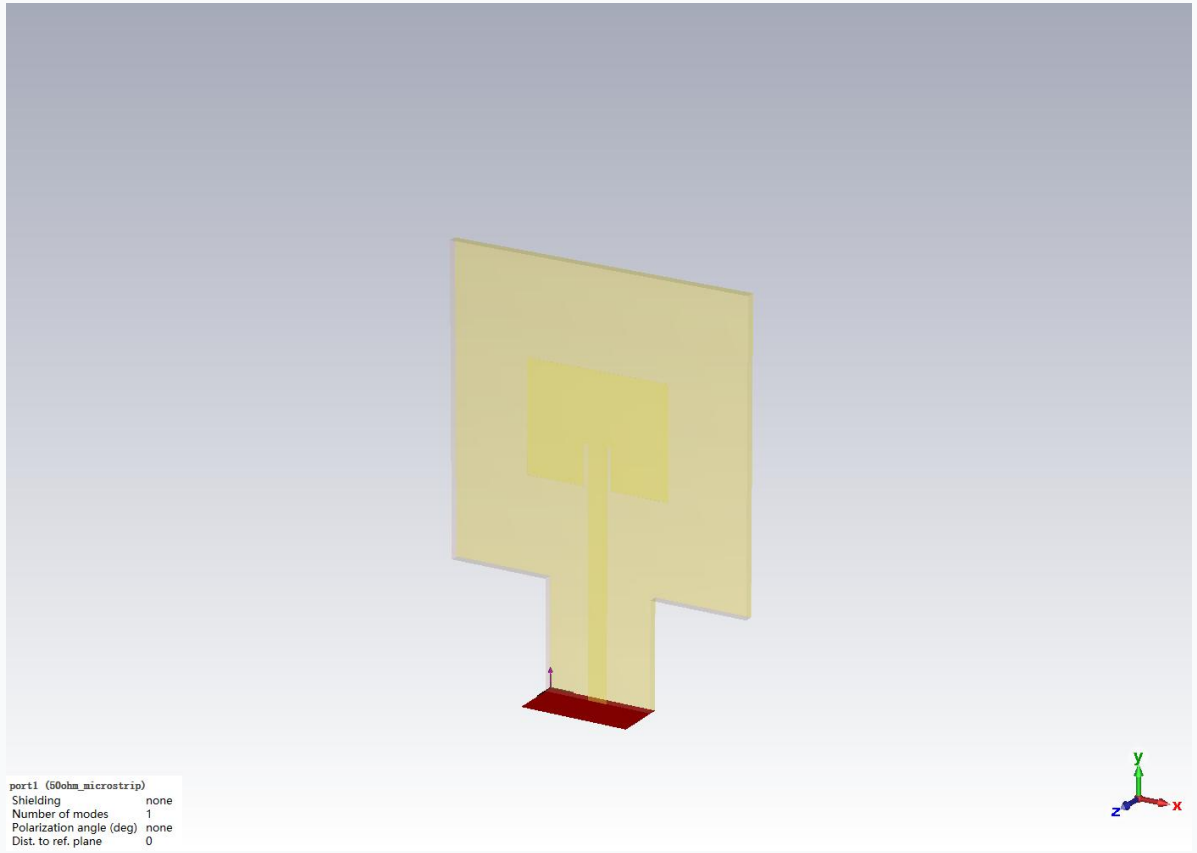
Final-size estimate: $c / [2(L + 2\Delta L)\sqrt{\varepsilon eff}] = 152.000$ GHz

Final geometry and materials

Patch W × L | 0.667 × 0.513 mm
Main ground / launch | 1.4 × 1.4 / 0.5 × 0.5 mm
Substrate | 0.040 mm polyimide; εr 3.5; tanδ 0.0027
Gold | 0.002 mm; conductivity 4.561×10^{7} S/m
Feed / inset / each gap | 0.090 / 0.190 / 0.020 mm

Material parameters are nominal D-band model assumptions.

Waveguide port and inset matching



Uniform launch: 0.5 mm long, feeding +y

Port plane: y = −1.2 mm; x = ±0.25 mm

z = −0.002 to 0.24 mm; one quasi-TEM mode

Final line impedance: 49.47 Ω at 150 GHz

S-parameter reference: 50 Ω

$R_{in}(y) \approx R_{edge} \cos^2(\pi y / L)$

Mesh, power and rerun checks

Coarse: 76,860 cells; 7.497 dBi; dip 149.680 GHz

Fine: 262,197 cells; 7.460 dBi; dip 148.840 GHz

Accepted 0.47727 W; radiated 0.34150 W

Metal 0.08954 W; dielectric 0.04264 W

Residual 0.00360 W = 0.754% of accepted power

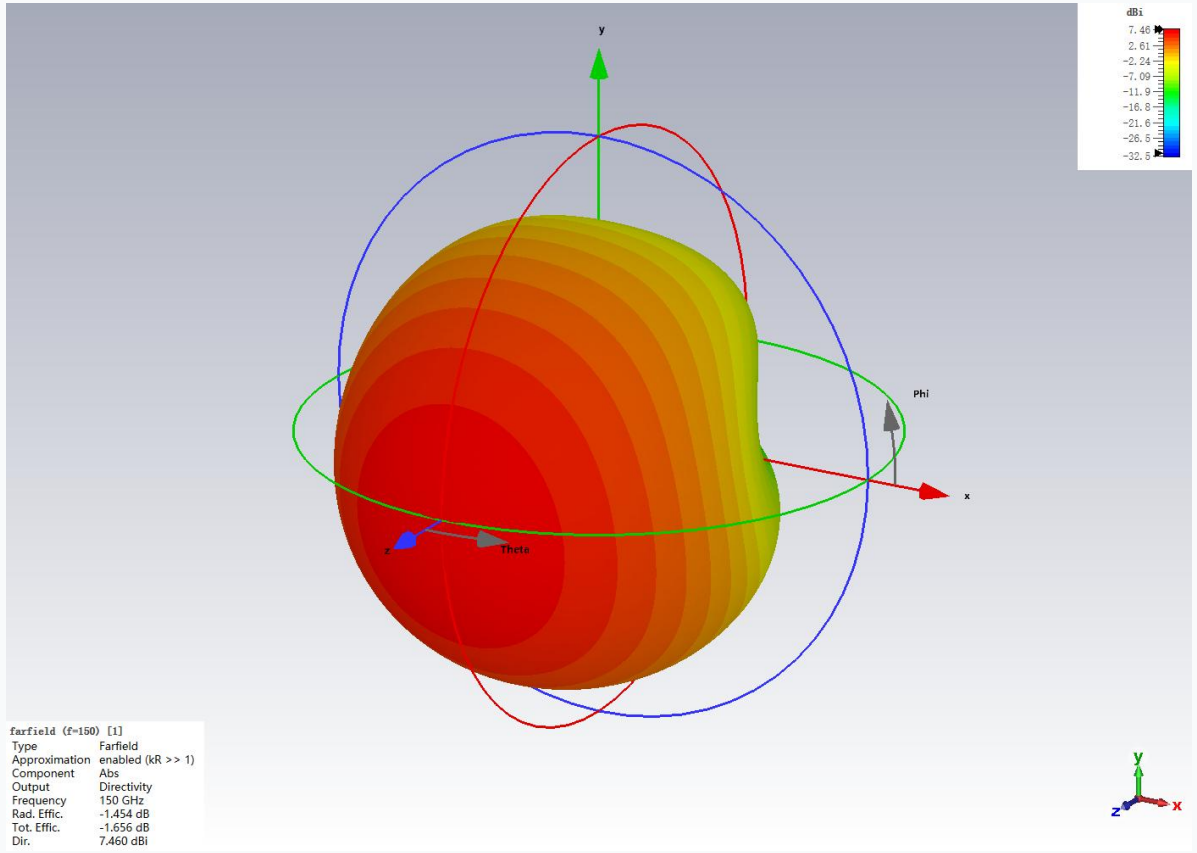
Spherical mean 0.9991; saved project rerun SUCCESS

Frequency results at all 13 monitors

GHz	S_{11} (dB)	Dmax (dBi)	Grealized (dBi)	ηrad (%)	ηtotal (%)
130	-0.71	5.14	-7.44	36.70	5.51
140	-1.58	6.55	-1.01	57.36	17.51
146	-6.43	7.21	4.58	70.72	54.62
147	-9.54	7.28	5.33	71.71	63.75
148	-15.91	7.35	5.82	72.12	70.27
149	-30.15	7.41	5.98	72.05	71.98
150	**-13.42**	**7.46**	**5.80**	**71.55**	**68.30**
151	-8.54	7.50	5.33	70.57	60.69
152	-5.97	7.53	4.66	69.04	51.59
153	-4.43	7.56	3.89	67.13	42.95
154	-3.44	7.57	3.09	65.06	35.62
160	-1.38	7.49	-1.08	51.03	13.90
170	-0.89	6.99	-4.70	36.57	6.77

13 actual farfield samples • 1001-point S_{11} trace on slide 7

Polarisation and 3D farfield



Peak at θ ≈ 4°, ϕ ≈ 264° on the 2° export grid

Broadside field predominantly y-polarized

E-plane yz: ϕ = 90° / 270°

H-plane xz: ϕ = 0° / 180°

Cuts on slide 9 show total directivity.

Diagnostic history and references

Invalid historical maximum: ≈ −20.92 dBi. Corrected port and uniform launch.

Near-model diagnostic: 8.54 dBi rejected; full-sphere mean 1.291.

Accepted: enclosing surface, −60 dB decay, mean 0.9991.

Theory: TELE9782 supplied project equations; C. A. Balanis, Antenna Theory, 4th ed., Wiley, Chapter 14.

Evidence: saved CST project, Iteration_Log.csv and Simulation_Validation.md.